Phase 1: Brainstorming & Ideation

Project Name: Rising Waters - Flood Prediction System

1. Problem Statement

Climate change and shifting weather patterns have significantly increased the frequency and severity of natural disasters, with floods being one of the most devastating. Traditional meteorological systems often struggle to provide localized, high-accuracy flood predictions in real-time. There is a critical need for a predictive system that can evaluate meteorological data quickly and accurately to issue early warnings, allocate disaster response resources, and potentially save lives.

2. Proposed Solution

A Machine Learning-powered web application that analyzes historical weather data and predicts flood risks. The system will leverage algorithms such as Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost to classify the probability and occurrence of floods based on features like temperature, humidity, cloud cover, and annual rainfall.

3. Target Audience

- **Meteorological Departments:** For validating and comparing ML predictions with traditional models.
- **Disaster Management Coordinators:** To foresee high-risk zones and allocate resources (boats, shelters, rations) proactively.
- **General Public:** To access an intuitive dashboard for current flood risk assessments in their local areas.

4. Key Features Brainstormed

- **Data Intake:** Ability for a user to input current meteorological parameters.
- **Predictive Engine:** A trained model (highest accuracy, likely XGBoost) to predict Flood Outcome (Yes/No) and Probability.
- **Web Interface:** An accessible UI (built with Flask and HTML/CSS) for easy interaction without requiring programming knowledge.
- **History & Monitoring:** Database storage of past predictions for continuous validation and monitoring.

5. Success Metrics

- Achieving a model accuracy of over 90% on testing data.
- Sub-second response time for prediction requests via the web interface.
- A clean, easy-to-use User Interface.